

Grounded Advisory Assistant (GAA)

Production-Grade Multi-Tenant Retrieval-Augmented Generation (RAG) System for Wealth Managers, Advisory Desks & Compliance Teams

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Architecture:	FastAPI + Groq + ChromaDB + Next.js 14	Compliance:	Zero-Hallucination & Tenant Isolation

1. Executive Summary

The **Grounded Advisory Assistant (GAA)** is an enterprise-grade AI decision support terminal built for private wealth management, relationship managers (RMs), and compliance auditors. Unlike generic LLM chat solutions, GAA enforces **strict grounded generation**: every response must be verifiable against bank-approved branch policy documents, fund factsheets, and regulatory circulars. If documentation does not support the query, the system triggers an automatic compliance refusal, completely eliminating costly financial hallucinations.

Powered by ultra-fast **Groq LLM inference**, tenant-scoped vector indexing, and real-time Server-Sent Events (SSE) streaming, GAA delivers sub-second response times with interactive, verifiable clause citation badges and slide-out excerpt inspection.

2. System Architecture & Tech Stack

Inference Engine	Groq API (openai/gpt-oss-20b / Llama3)	Ultra-low latency streaming, 100% cost-effective cloud execution.
Vector Storage	ChromaDB + Custom Deterministic Embedding	Clause-level chunking, Cosine similarity, strict metadata filtering.
Backend API	FastAPI, SQLAlchemy, SQLite/PostgreSQL	Async SSE token streams, multi-tenant isolation, PBKDF2/JWT auth.
Frontend UI	Next.js 14, React, TailwindCSS, Lucide Icons	Enterprise dark UI, interactive citation drawer, persistent chat history.

3. Core Features in Short

- **Zero-Hallucination Grounding:** Every answer is synthesized strictly from retrieved policy documents. If the similarity score is below the threshold, the system triggers an automatic compliance referral to Legal & Compliance.
- **Interactive Verifiable Citations:** Direct clause citations (e.g. [Doc: Policy_2024, Clause: Sec. 4.2]) rendered as clickable badges that open a slide-out drawer showing page number, similarity score, and verbatim excerpt.
- **Multi-Tenant Branch Isolation:** Complete logical separation between bank branches/accounts. Relationship managers can never retrieve, query, or view another branch's internal policies.
- **Fast Groq Cloud LLM:** Integrated directly with Groq's high-speed inference engine, providing rapid streaming responses without requiring OpenAI credits or paid subscriptions.
- **Document Knowledge Base Store:** Allows branch admins to upload, parse, and vector-index PDFs, Word documents, and policy circulars with automatic chunking and metadata extraction.
- **Compliance Audit Trail:** Complete logging of queries, rewritten standalone queries, retrieved chunk IDs, similarity scores, latency, and refusal decisions for regulatory reporting.
- **Persistent Chat History:** Robust client-side session management preserving conversation context across page navigation between Advisory Chat and Document Store.
- **Role-Based Access Control (RBAC):** Distinguishes between Relationship Managers (RMs - advisory queries) and Compliance Admins (document management, audit review, and deletion permissions).

4. Security & Attack Simulation Verification

WealthGuard GAA features a dedicated, automated attack simulation and SLA verification test suite (``pytest``) designed to validate zero-hallucination compliance and multi-tenant security barriers under hostile conditions.

Cross-Tenant Breach	User from Branch A requests private wealth policies stored in Branch B.	PASSED: Vector query metadata filter restricts search exclusively to Branch A <code>account_id</code> .
Hallucination Injection	Adversarial query asking for unverified tax exemption loopholes not in doc store.	PASSED: Grounding threshold triggers Refusal Protocol; zero invented terms emitted.
Prompt Injection / Jailbreak	Prefix instructions attempting to override system instructions and leak raw prompts.	PASSED: LLM prompt isolation shields system guardrails and context structure.
Unauthorized Document Tamper	RM role attempts to call DELETE /documents endpoint.	PASSED: HTTP 403 Forbidden enforced via FastAPI RBAC dependency.

5. Step-by-Step Testing & Execution Guide

Step 1: Backend Execution

Navigate to the ``backend/`` directory, configure your ``.env`` file with your ``GROQ_API_KEY``, and start the FastAPI ASGI server:

```
cd backend
python -m uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload
```

Step 2: Frontend Execution

Navigate to the ``frontend/`` directory, install packages, and launch Next.js in development mode:

```
cd frontend
npm install
npm run dev
```

Step 3: Running Automated Security & Unit Tests

Execute the complete comprehensive automated test suite (including tenant attacks and SLA verification):

```
cd backend
python -m pytest tests/ -v
```

Ready for Enterprise Production: WealthGuard GAA delivers a complete compliance-hardened advisory interface. All OpenAI dependencies have been cleanly replaced with Groq API key support, and chat history persistence across the Document Store is verified and stable.